Architectural Design and Discourse Studio 2023
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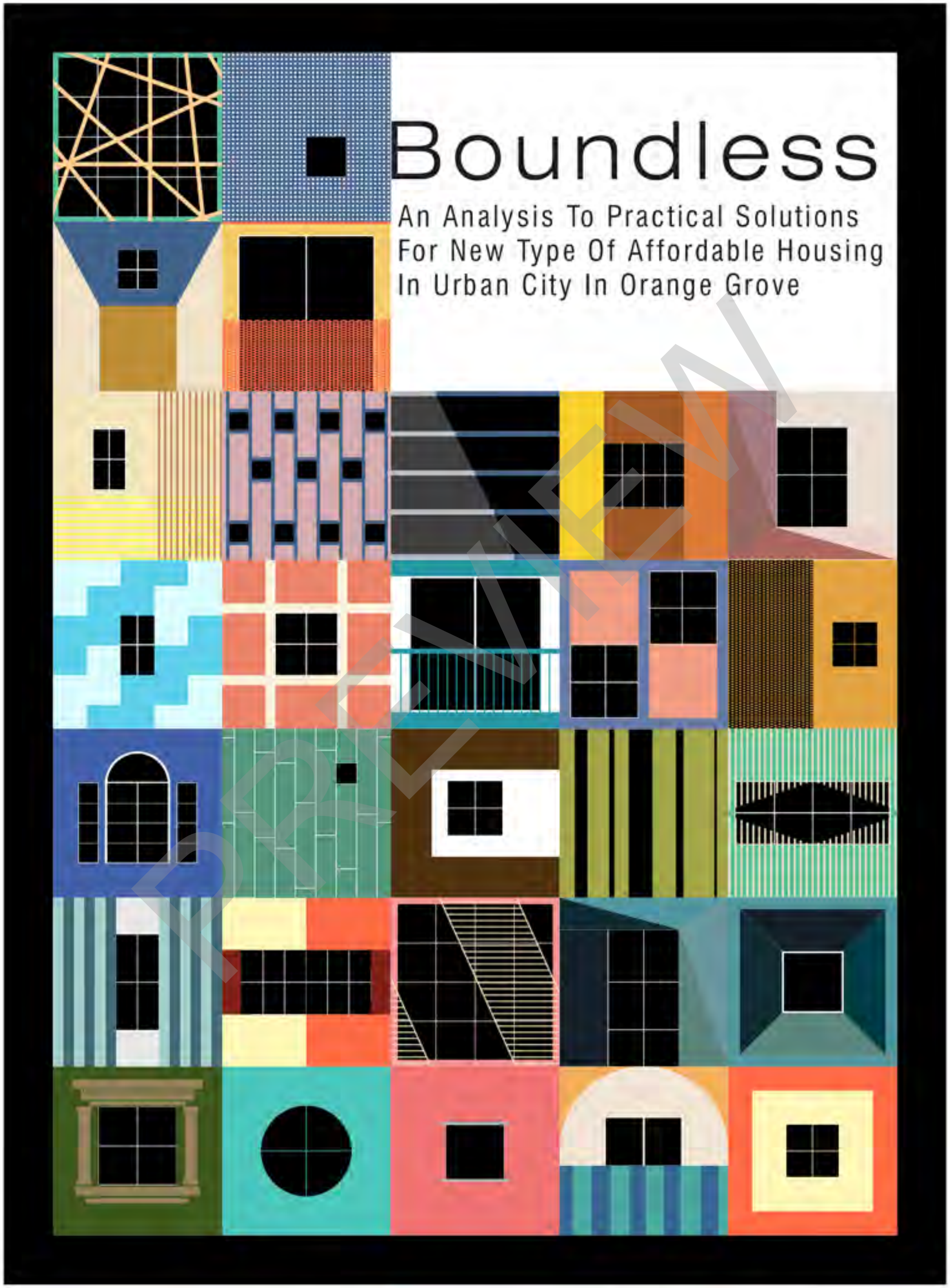

This is a research report submitted in partial fulfilment for the degree
master of architecture (professional) at the Univerisity of the witwatersrand,
Johannesburg, South Africa, in the year Feb 2023

Declaration

I, Zhihao Zhang 1665340 am a student registered for the course Master of Architecture (Professional) in the year 2022.

In accordance with Regulation 4(c) of the General Regulations (G.57) for dissertations and theses, I declare that this dissertation, which I hereby submit for the degree Master of Architecture (Professional) at the University of the Witwatersrand. There is no part of my dissertation has already been, or is currently being, submitted for any such degree, diploma or other qualification. I further declare that this dissertation is substantially my own work.

For this purpose, I have referred to the Graduate School of Engineering and the Built Environment style guide. I hereby declare the following: I am aware that plagiarism (the use of someone else's work without permission and/or without acknowledging the original sources) is wrong. I confirm that the work submitted for assessment for the above course is my own unaided work except where I have stated explicitly otherwise. Where reference is made to the works of others, the extent to which that work has been used is indicated and fully acknowledged in the text and list of references.

Zhihao Zhang
15-Feb-2023

This document is submitted in partial fulfilment for the degree: Master of Architecture (Professional) at the University of the Witwatersrand, Johannesburg, South Africa.

Acknowledgements

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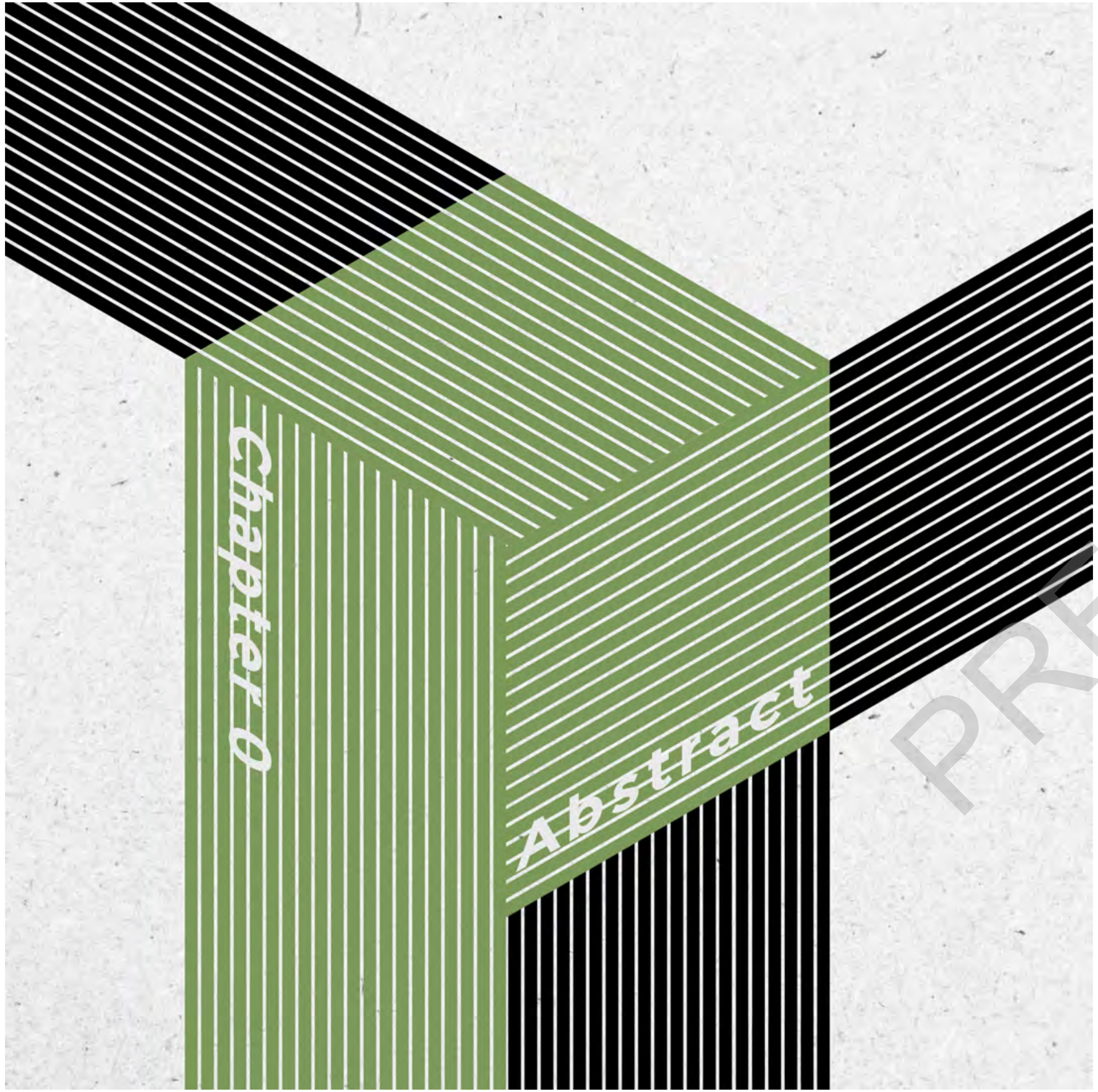
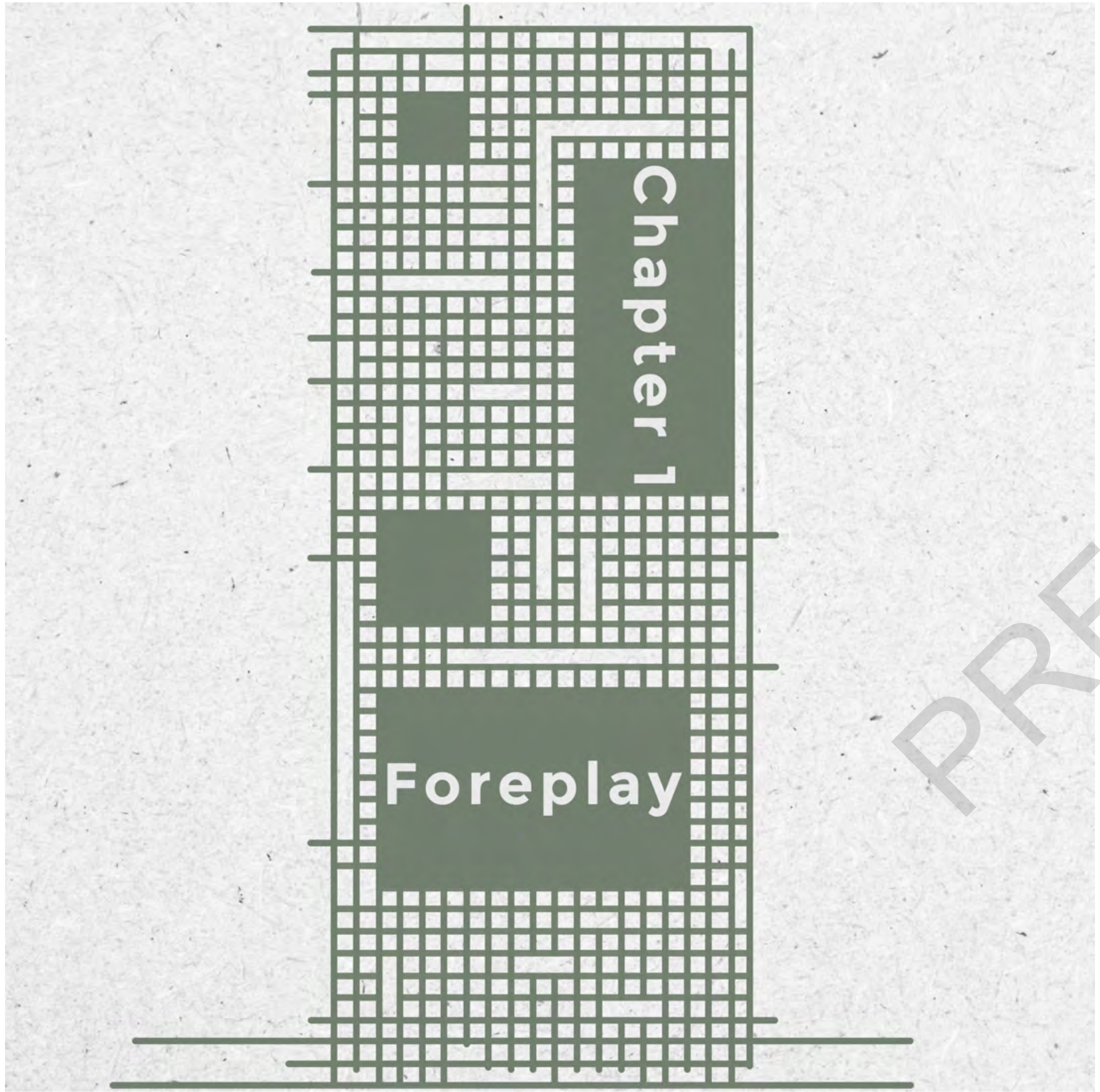


Abstract

The research aims to solve the issue of low-density and urban sprawl in Johannesburg focusing on Orange Grove as catalysis for higher density. The majority of working-class citizens still live outside of the city, commute daily at a significant cost, and travel far distances to access work and economic opportunities. Research shows that 60% of Gauteng commuters spend more than 10% of their income on transportation alone (GSIR, 2019).

As a result of high commute costs, many people are less productive, remain financially insecure, and have less time to spend with their families. The proposal is to create sustainable communities with a unique goal, ensuring secure and low-cost housing for the mid-to-low income class.

There are many South Africans who need proper housing that offers a sense of belonging, security and privacy, but live in difficult and uncomfortable conditions in informal settlements. Urban areas are also experiencing strain due to economic factors and rapid urbanisation, on top of the backlog of housing demand. The idea of high-rise apartments will provide housing for working-class people close to public transportation and area development. This proposal also investigates how a high-rise solution may be used to address the economic, socio-political, housing, and sustainability issues that come along with the densification of an urban area.



Introduction

All people should have a roof over their heads as a human right, specifically in South Africa, one of the few countries with the right to housing included in our constitution (OHCH, 2021). Unfortunately, it is a sad reality that many families in South Africa struggle to afford this necessity, particularly in mid-to-low income areas or single-family homes. Housing shortages have become an ongoing problem in many mid-to-low income neighbourhoods, especially in Johannesburg (John Campbell, 2017).

According to the Department of Human Settlements, there is an estimated backlog of 2 million houses. It might seem like a large number, but consider, on average, six people in each family, leaving about 12 million people who need housing (Harsch, 2020). Despite wanting to save for retirement, most middle-class South Africans typically spend their remaining income on repaying debt after paying for essentials (Harsch, 2020). Still, money for the barest living essentials, economic pressure and social instability add to the financial stress of the mid-low income family unit.

Figure 1: Sketch of view of informal settlements in Alexsandra, sketch by Zhihao Zhang

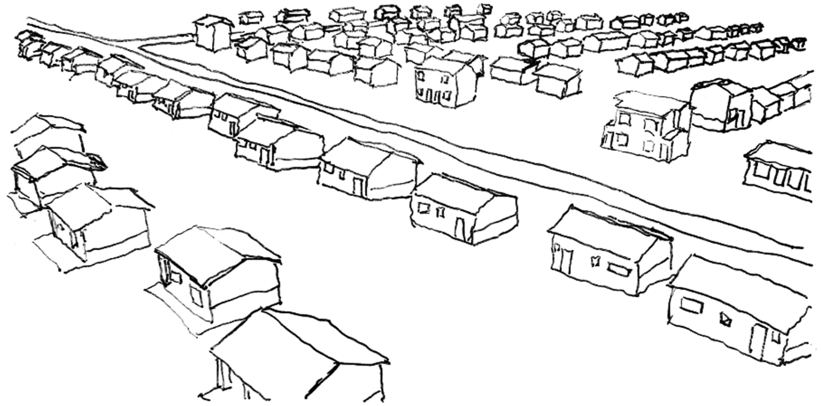


Figure 2: Sketch of subsidised housing backlogs in Westonaria, sketch by Zhihao Zhang

Human activity and interaction have become increasingly important in cities due to population growth and rapid urbanisation. (The process by which a rural community may become part of a larger urban area can be described as suburbanisation. (Coolgeography, 2022)) Suburbanisation is the result of the already densified city being unable aid the current and increasing population. Johannesburg's epicentre of economic activity and sophisticated infrastructure is far from the high-density development of Johannesburg. Due to this, the city has enjoyed a dual development trajectory with high-density development in the periphery, and low-density development in the economic core. (Joburg Newrooms, 2018) In order to meet the needs of today's population without compromising tomorrow's, researchers are working to develop a sustainable form of city life.

Compressed cities, specifically sustainable compact cities, are growing due to increasing economic and socio-political needs during the 21st century. To keep up with growing international trends and increasing awareness of sustainable practices. By reducing energy consumption and building urban elements in an efficient and compact manner, we can revitalize the traditional city centre, which has declined due to suburbanization. (Nallathiga, 2007) These actions could contribute to combating global warming. The best way to achieve a compact city form is to centralize the various functions of the city in high-rise buildings.

High-rises and sustainability are intrinsically linked. (Mills & Bachand, 2018) There are four values represented by high-rises: rivalry, originality, power, and affluence. (Cherkaoui, 2018) In contrast, the sustainability concept emphasises long-term strategies and mutual gain as a means of maximizing diminishing resources. Described as reducing resource consumption and waste while improving liveability in urban areas, sustainability is defined as reducing resource consumption and waste. A sustainable approach ensures minimal impact on the environment and the users during design, construction, and maintenance.

It is essential that there is enough supply of sustainable housing, that it is built adequately and sustainably, and is affordable, with the possibility of providing for the initial costs and exploitation expenses for numerous households. An ideal home must encompass aspects such as energy efficiency, eco-friendly construction materials, environmentally friendly waste management, aesthetics, comfort, and a sense of home, reflecting a

social-psychological aspect that meets a person's requirements (Marco, 2017).

It could be generally understood that liveability encompasses the factors that contribute to the quality of life, including safety, economic opportunity, health, convenience, mobility, and recreation in the home, neighbourhood, and metropolitan area. Accordingly, the liveability of a nation is measured by how well it provides for its citizens compared to their needs and capabilities. In all definitions, liveability reflects a person's quality of life, well-being, and satisfaction of their needs - and all definitions are inherently anthropocentric (Spinelli & Lionetti & Pastore & Fasolo, 2020).

It is the symbolic value of the high-rise buildings for the communities in which they are located that is one of their greatest attributes. Economic growth is symbolised by high–rise buildings, which can become landmarks and contribute to residents' pride and satisfaction (Daniel, 2022). As technology and society develop, symbolism depends on their relationship. However, social processes can be complex and difficult to regulate. A key aspect of human well-being is maintaining biodiversity in ecosystems. According to research, (Fabinyi, Evans & Foale 2014) humans seek to retreat to environments similar to those that surrounded our evolution to interact with nature. It is a matter of well-being and sustainability for residents to be compatible the nature of a high-rise.

Due to its establishment during the Gold Rush and the legacy left behind by Apartheid (Trangos & Bobbins, 2015). During the gold rush, city freeways, settlements, and ecological systems are awkwardly intersected by a landscape of extraction. In order for the city-region to develop and integrate spatially in the future, the uninhabitable mine waste pockets that remain present challenges (Luzipho, 2014). Taking a cross-section of Johannesburg, running north-south, The city transitions from a tree-lined suburb once shaped by a rich mining master, containing decentralized economic nodes, to an inner city dominated by the concrete and vertical old central business district. Within the inner city, light industry, manufacturing, and warehouses are interspersed with the remains of mining activity. As a result of urban lifestyles, industry, and mining, these essential infrastructures and resources are constantly under threat. It is located away from social, cultural, and economic centres and many high-density areas emerge on the periphery of the city region, including Kwa Thema, Katlehong, Ga-Rankuwa, Kagiso, and Orange Farm (Bobbins, 2015). These rural suburbs are spatially disadvantaged due to their dependence on unreliable public transportation (Trangos & Bobbins, 2015). Johannesburg's unique urban fabric and high-rise buildings make this country an intriguing place to conduct research. The Freedom Corridor Project was conceived in Johannesburg in the context of this multi-layered and continuous legacy of apartheid (CITY OF JOHANNESBURG, 2018). As part of the initiative, several pre-existing areas will be connected via a large transport corridor that will be linked to interchanges centred around which mixed-use developments and increased accommodation density will be focused. By creating public spaces, programs, participation, and ecological architecture, it aims to increase gratification for users and create a sustainable community.

Research Questions

How does Johannesburg's urban planning affect the quality of life, and what is the balance between Johannesburg's future development and past legacy?

Landscape, society, and ecological systems in Johannesburg have been fundamentally changed by the legacies of extraction. The golden mining period has also undoubtedly contributed to the emergence of distinct racial divides during the city's formative years. As a result of this, in conjunction with the planning strategies of apartheid, Johannesburg's urban form became fragmented, the natural landscape altered, and spatial integration hindered.

One must reflect on the relations of the past to completely grasp the current context surrounding Johannesburg, associated area development and communities located along its boundaries. There is an inextricable link between the history of heritage conservation and past city planning presence. As the government and local communities collaborate toward reconnecting people with the environment by conserving and sharing benefits as common goals, while understanding and respecting the history of the community.

Is there a way for architecture to promote growth-supporting social interactions within Johannesburg that will grow and intensify over time?

The development of a conceptual framework which links the quality of life to passive sustainable design responses could be investigated. A second response would be an analysis of high-rise buildings in terms of their sustainability and quality of life to produce a theoretical output.

Architecture should play a role in fostering social interactions on site (tectonic), this being key to the success of my proposed architectural intervention. It is important that, due to the complexity of Johannesburg, the final intervention responds responsibly to its surrounding environment and draws attention to areas throughout. The site's surrounding factors must be thoroughly understood to achieve this.

Research Methodology

Creative research through design is a process that combines inductive reasoning, interpretation, and imagination with various themes, constraints (like technological constraints, the building site, clients, and users), and precedents (urban and architectural). To achieve its creative outcomes, creative research as a discourse-based and design-based process is iterative (inductive-deductive), cumulative, cyclical, and open-ended. As a result, knowledge and argumentation are derived through qualitative as well as quantitative approaches, while the creative approach is used to translate the same into a defensive architectural proposal.

The thesis will be predominantly written using books and Internet resources like those described above for the majority of the research.

There's also personal communication and site-contextual analysis that will be supported by the thesis.

Contextual analysis of the site examines the current conditions so that appropriate responses can be incorporated when designing the building of the site to incorporate social and environmental influences. The use of personal communication, including email and informal discussions with experts, will be very useful in obtaining information about high-rise apartment building construction and design principles. In order to understand what middle- and low-income groups want and can afford in an apartment building, the thesis needs to interview middle- and low-income groups.

Aims & Objects

There are four objectives of the study, including strengths, weaknesses, opportunities, and threats related to high-rise buildings' impact on quality of life:

1. Assess the relationship between sustainability and quality of life indicators in high-rise buildings

2. Analyze the liveability of high-rise buildings to improve their quality of life.

3. Assess the impact of energy-efficient solutions on enhancing the quality of life.

4. Identify and assess the impact of the design of high-rise residential buildings on the immediate and broader urban/ city environment.

The objective is to explore how architecture impacts the quality of life through design. It is important that architects determine whether high-rise residents are satisfied, if their needs are met, and if their quality of life is adequate. With a commitment to social development, architects have worked to end racial and class segregation, which places affordability and quality of design at the forefront of the design process.

Although the boundless concept referred to in the title and throughout this thesis is a new form of living in vertical cities, and provides security and safety to its inhabitants, the intervention seeks to make this more porous so that there may be greater opportunities for economic growth and capacity building in Johannesburg. The architecture will be adapted to the site, context, and climate of the current site to achieve this. To achieve long-term sustainability and ecological balance, architecture must satisfy both human and environmental needs. The design should be environmentally sensitive and socially relevant but also engage and empower residents to become more aware of their surroundings.

The Proposal For High-rise Housing

The rapid growth of the human population and the shortage of non-renewable land resources have become the greatest contradiction and the greatest threat to human society from the point of view of human habitation. Even though birth rates have declined significantly around the world, it is believed that most of the world's population will increase in the future (Rahul Mittal, 2013). The contradiction between the rapid population growth and the shortage of land resources in modern urban development must be resolved, as well as planning for the next generation in the next 50 years, leaving enough room for development (National Planning Commission, 2022).

From the perspective of environmental architecture, there are two indicators to solve the question of housing more people on the same piece of land: the density of the buildings and the volume of the buildings (Iman, 2020). Housing developments in high-rise buildings can be a practical solution to this problem.

Despite what we see today, high-rise buildings are not the first or only tall structures built by humans. Whether they were built for different purposes or symbolic purposes, the pyramids of Giza, ancient church towers, and the Eiffel Tower all shared a common characteristic of being tall structures. However, they still serve their respective purposes even though they aren't the most practical or economic buildings.

Building a high-rise can be motivated by several factors. A building's driving force determines what is central to its design, so understanding what it is is critical. In contrast to creating a new icon in the city, building as economically as possible is a distinctly different approach (Mir & Kheir, 2012). As land prices increase and people choose to live close to city centres, high-rise buildings are developed to meet this demand. Large cities such as New York and London, where land value is drastically high, are a plausible explanation. New York's Flatiron building is a prime example of where high real estate prices made it difficult to build on an economical plot (Edward Glaeser, 2011).

Creating a denser city is another reason for building high. As a result, more people can live near their workplaces and amenities, thereby decreasing the need for transportation. Sustainable lifestyles are made possible by it. Public transport is also efficient in high-density areas due to their high population density.

Creating a new iconic building is also a common goal. The icon may represent a country, a city, an organisation or an individual (Khaled, Dina, Mohamed, 2020). It is common for cities to compete to build taller buildings than their rivals in many regions. Not only does an iconic building get publicity, but it can also serve in marketing and symbolise a city's power.

Figure 3: Watercolor Drawings of the City of the Future , drawing by Maja Wrońska, 2017

High-rise Housing: The Liveability Controversy

Despite the benefits of high-rise housing, one of the main arguments against high-rise living is liveability. Nevertheless, a new "high-rise era" has opened with the promise of energy efficiency, land conservation, and sustainability. According to Liveability, the environments are evaluated based on their immediate needs and experiences, including the health and well-being of their occupants, as well as their immediate quality of life, looks at the role liveability plays in contemporary urban theory and practice and contends that sustainability needs liveability (Feliciotti, 2018).

Sustainability must also consider liveability, which focuses on the immediate needs and experiences of residents, their quality of life, and the environmental and health impacts of their choices. If these are achieved at the price of liveability, they will inevitably face resistance in the real world. In this connection, sustainability and liveability are complementary and cannot be achieved without the other. As a result of this theoretical understanding, the liveability of high-rise housing is of considerable research importance and practical significance, while offering sustainability benefits (Iman 2020).

Research conducted about high-rise residents found that families raising children tend to want to live on the ground level (Wolff, 1987). The results showed that the environmental characteristics of high-rise housing present certain liveability issues, including the fact that residents must share space with strangers. It also found that communal living leads to conflicts between residents. Other negative factors such as noise, pollution, poor security and overcrowding came into play. From the research, the development of high-rise housing has always been controversial. These arguments claimed that these issues caused the high-rise housing crisis. (Gifford, 2007) Despite technological advancements, the quality of life in high-rise housing continues to improve. But still as many researchers continue to argue that while high-rise housing may offer some benefits, these benefits are not enough to outweigh its negative effects on its occupants.

Problem Of High-rise Liveability

Negative effects on the family raising the child

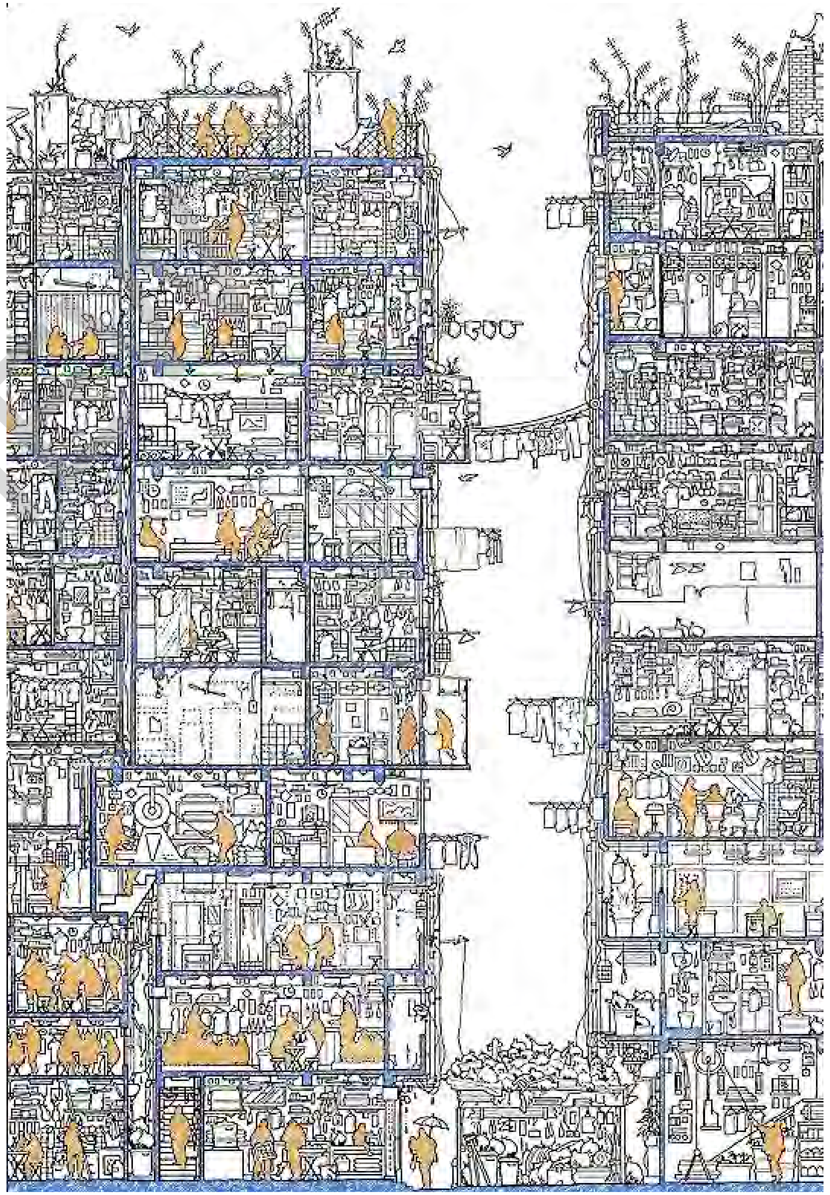


Figure 4: Future Living Inspired by Past Extremes - Kowloon Walled City & Hashima Island , 2010

The liveability of high-rise housing was debated initially based on the effects that living in a high-rise had on young families. According to several studies, high-rise housing can negatively impact families, including increased conflict between family members, parental isolation, and behavioural and developmental disorders in children (Spinelli & Lionetti & Pastore & Fasolo, 2020). There is a distinct difference between high-rise housing and other types of housing in that children's outdoor space is closely linked to the home. The children in most high-rise residential areas are usually only allowed in public (in open communities) or semi-public (in gated communities) areas, which are often outside of the direct supervision of parents. Customers who live in these high-rise residential areas often keep their children indoors for fear of not being able to supervise them, which has many negative effects.

The children and parents both suffer from the lack of access to outdoor spaces. The lack of physical activity and group activities can lead to certain physical and psychological difficulties in children. Research has shown that children living in high-rise housing usually begin to play independently outdoors at an older age than those living in low-rise housing, and they are less likely to communicate with their peers (Sandstrom & Huerta, 2013). Due to fewer physical activities and outdoor exercise, these children experience a lag in their physical development, which in turn leads to respiratory problems and other behavioural problems and causes lagging development in children on high floors. Additionally, the fact that the place of activity is often out of sight of the parents can lead to problems of under-supervision both in the outdoor and indoor environments.

The difficulty of raising children in high-rise housing, especially on higher floors, is also apparent from the parent's perspective. High-rise living can increase the level of stress experienced by parents due to the lack of a safe place for their children to play. It found that the stress increased as the floor rises. At the ground level, children are cut off from sight and sound communication with their parents, which can negatively affect their psychological well-being. Mothers living in high-rise apartments were more likely to suffer from depression than those in low-rise apartments. The high density and poor security of high-rise apartment buildings left mothers feeling isolated and more likely to experience social isolation. As a result of high-rise living, parents must frequently keep their young children at home, a restriction that contributes to conflict within the family limits opportunities to play and reduces interaction with neighbours.

Figure 5: The photograph of Kids Live in Kowloon Walled City , Hong Kong, Confinement by Khanh Nguyen

Medical conditions for residents

Studies on high-rise living indicate that it can negatively impact a person's mental and physical health. People who live in high-rise buildings can lead to breathing problems, headaches, and bouts of illness, known as "Sick Building Syndrome (SBS)". (Uwaegbulam, 2018) High-rise housing residents' mental health indicators were found to be deteriorating, while an increasing number of residents were seeking medical treatment for psychological problems. High-rise housing can lead to negative effects like mental stress and social isolation, as well as an inability to avoid the interference of different lifestyles and habits, which interferes with the separation between people. There are also no restorative effects of nature and landscape on residents because of living in high-rise housing.

The research found that mortality rates declined with the number of floors occupied, with those living on floors eight and above being 22% less likely to die prematurely than those on the ground floor, and 40% less likely to die from severe heart and lung disease. Furthermore, such a result might indicate that people who tend to be less healthy living on the lower floors. (Uwaegbulam, 2018)

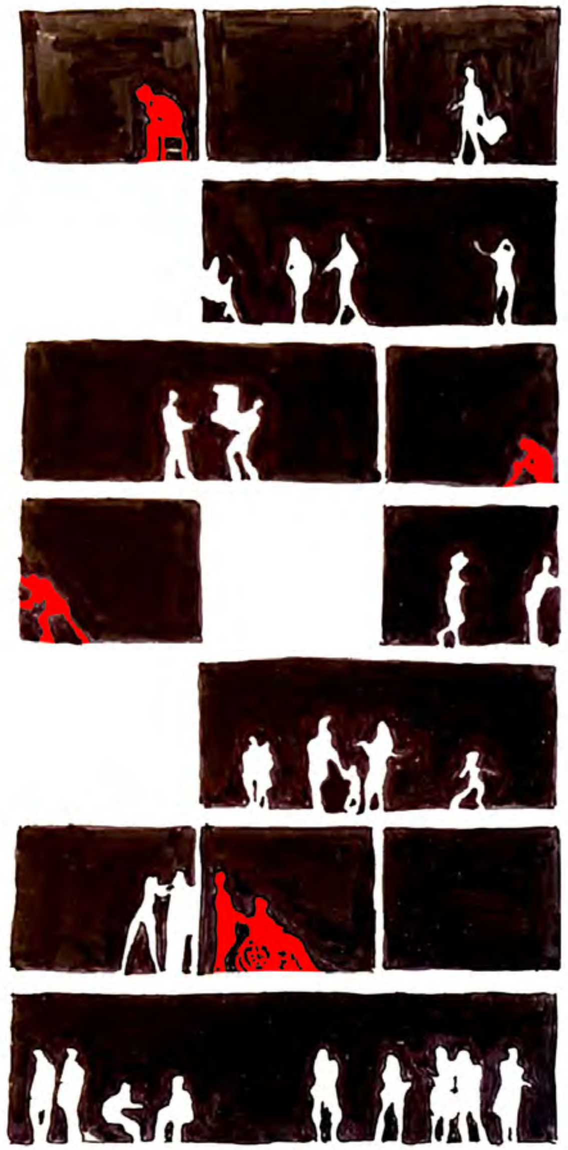


Figure 6: The sketch of section of high rise living quality: number of residents get mental stress and social isolation, drawing by Zhihao Zhang

Unsafe conditions for residents

Figure 7: The sketch of an accidental fire in high rise, drawing by Zhihao Zhang

Since most residents of high-rise apartment buildings do not live on the ground, security may start to present problems. It was found that high-rise residential living is prone to causing feelings of fear and insecurity due to factors such as fire, natural disasters, communicable diseases, and equipment failure (such as elevators, water supply, and power supply failure) for current and potential residents. (Ronchi & Nilsson, 2013). For these reasons, many people fear living in high-rise buildings. Although fire safety and equipment failure could be improved through technological means, the fear of the residents cannot be eliminated. Adapting to this new way of living will take time.

- Poor social communication among residents

Studies have found that residents of high-rise residences are affected by social contact from two perspectives: relationships with neighbours and a sense of community. Nearly every urban individual in contemporary society engages in informal relations with their neighbours on a daily basis (Ruonavaara, 2021). According to a survey, the individuals in a residential building are likely to feel isolated, finding people who live upstairs and downstairs have far more understanding of others than people who live on a street in terrace housing (Yao, 2020). The research showed that people living in high-rise residential buildings maintain fewer relations with their neighbours than those in low-rise condominiums. The residents of high-rise buildings have more neighbours living in the same building and have more acquaintances, but more contact and interaction between them than the residents of low-rise buildings. Interaction among residents of the upper social outdoor and indoor living areas is considered less intense than that of the low-rise residential area, according to some studies. Overall, high-rise residential homes are often characterised by a lack of understanding among residents, a low level of social support between neighbours, the deterioration of public space and social interaction, and the adverse impact of social alienation of residents.

In general, community consciousness includes a sense of belonging and identity, a low level of community consciousness will affect the residents' quality of life. Those who live in high-rises are more inclined to have colleagues or classmates living in the same region than those who live in the same building. It is critical to be aware that poor social contact can also result in individuals having higher levels of privacy and will not need to interact socially with anyone.

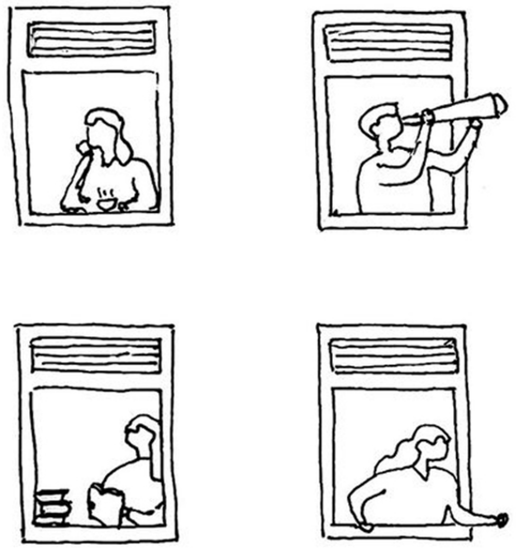


Figure 8: The sketch of poor communication with neighbours in high rise, drawing by Zhihao Zhang

- Deterioration of safe conditions in the high-rise community

One of the major problems with high-rise buildings is that their security is questioned and criticised by scholars, and as a result, they increase crime rates and lead to such antisocial behaviours as vandalism and graffiti. Initially, residential building was considered a type of public housing for working-class families, which resulted in the excessive concentration of low-income families in residential areas, which worsened the social environment.

As soon as there are other housing options available when quality problems occur in large-scale high-rise housing, people start to "choose with their feet" and are replaced by those who have no choice, causing poverty to spread and the residential environment to degrade further, forming a vicious cycle (filtering down effect), high-rise residential buildings are likely to become the new urban slums if timely intervention by external factors isn't made. When it comes to housing policy and management, scholars have suggested effective solutions to this problem, the building system should have social diversity and strict management control as effective ways to improve the public security environment of the high-rise housing complex.

Figure 9: The low-income group lives in an old apartment in Hanoi., image by Photo: Quang The / Tuoi Tre, 2021

Some scholars, however, believe that high-rise residential buildings exhibit some built environment characteristics that contribute to poor public security conditions, such as the increasing amount of communal space (entrance halls and elevator halls) and the facilities (elevators and stairs) that cannot be effectively supervised by residents, which can be attributed to the anonymity of residents, as well as the "pro-crime" environment with elevators, poorly lit corridors, bicycle storage areas, laundries, and, most importantly, underground parking lots (Dennis Challinger, 2008).

There is a risk that a property company or relevant organisation cannot manage and maintain the property effectively, leading to litter, graffiti, and criminal activity. The high-rise residential building crime rate was significantly higher than the low-rise residential building crime rate, in particular the crime rate in public areas of high-rise residential buildings was much higher than that of low-rise residential buildings.

There are a few possibilities - firstly, due to high-rise buildings having similar locks, burglars do not have to search for the right target as they have the right picks nearby. Second, due to the large number of households living in one building, high-rise buildings may lack territorial and natural security. Third, occupants may not have strong social ties (Shimada, 2004). The building design could be suggested by rules and design methods, such as enforcing residential boundaries, monitoring buildings and residential entrances 24/7, and strengthening and encouraging resident supervision awareness, to improve high-rise residential security.

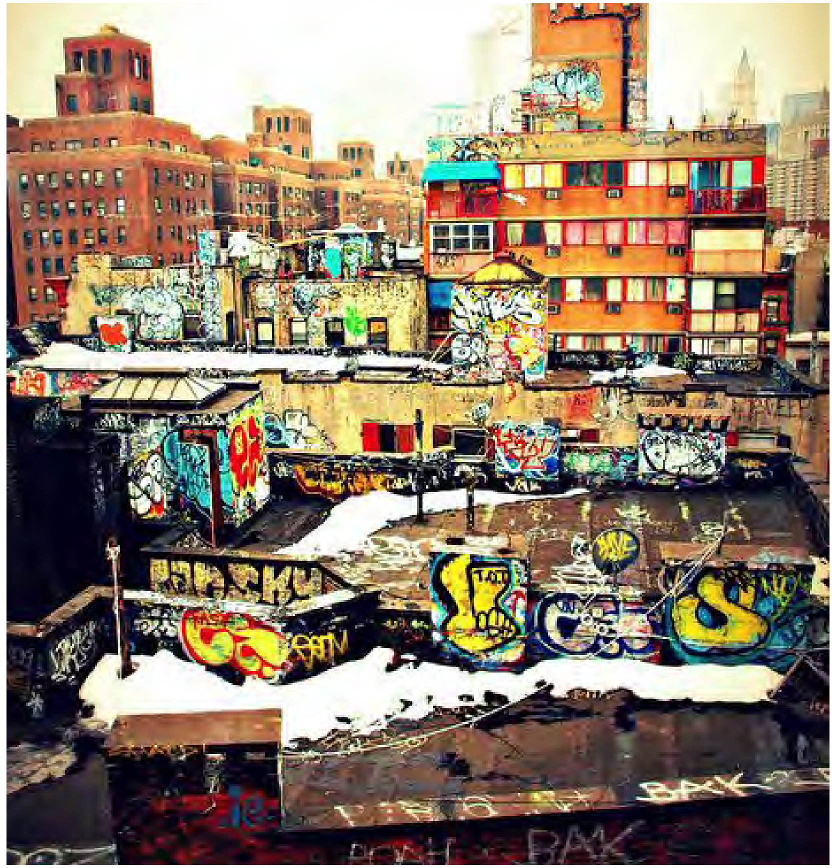
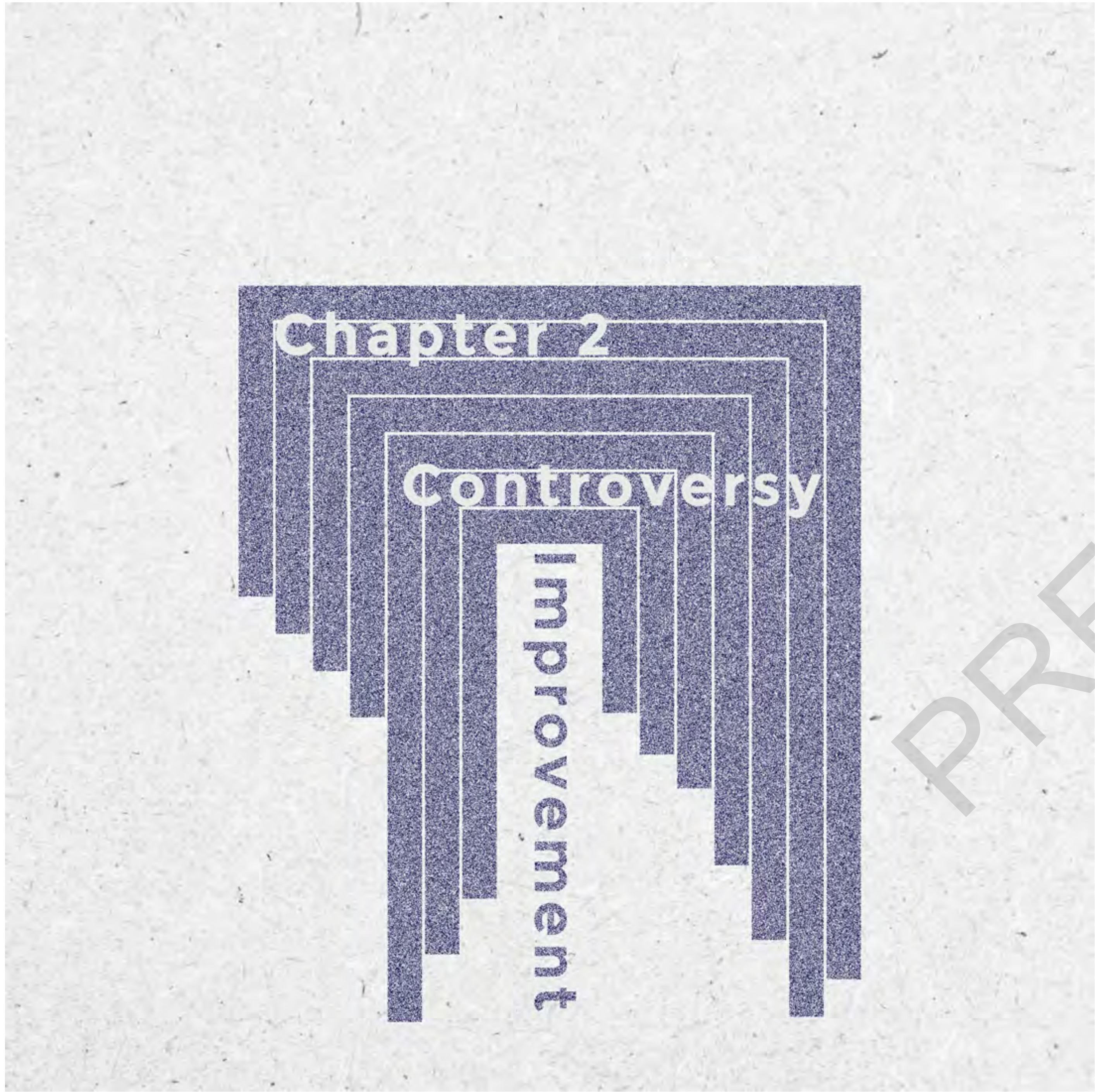


Figure 10: The photo of graffiti on the buildings, Photo by NYCUrbanScape, 2011



The History Of High-Rise Building

The Origin of high rise building

It is only recently that the human experience of living has been metaphorically, and literally, elevated to the same level as the tall structures once reserved for religious or cultural buildings. Throughout history, humans have continuously attempted to build tall structures, driven by religion, culture, and faith. The pyramids in Egypt, the pagodas in China, and the steeples of European churches all represent different civilisations. However, until the twentieth century, the dwelling was rarely built very high, being the most dominant form of human construction (John Alfred Thomas, 1994).

As one of the first group of architects who designed skyscrapers, Louis Sullivan wrote that " form ever follows function, and this is the law", the function of the house depends on the essence of living (Jackie Craven, 2019). A notable characteristic of the history of residential evolution is the way that the basic traits of residential patterns change over time, and how they reflect human basic living requirements of stability and continuity.

Figure 11: Pyramids, pagodas, steeples at sunset, the photo by Ash Edmonds, JIM QUINN, Digital Vision

Appearance: A modern housing type based on functionalism

The first high-rise building was built in Chicago in 1884. It is called the "Home Insurance Building". This building was designed by the American architect known as the father of the skyscraper, William Le Baron Jenney (Architectuul, 2022). High-rise towers, influenced by the Chicago School of Architecture, began to appear in Europe and The United States around the time of the invention of pre-stressed concrete, high-speed elevators, pressurised pumps, air-conditioning systems, and flush toilets.

After World War II, the German economy was impacted by the reconstruction and labour movements (Silesian weavers' uprising, 1844), and the architects sought to build a cost-effective form of housing. Traditional low-rise garden houses were replaced by densely packed multistorey and high-rise apartments. Technical standards were developed, such as minimum square feet per person and floor spacing, to ensure that each house would receive ample sunlight, fresh air, and green space. The house planning layout was called "Zeilenbau" (Eric Mumford, 2019),

***"... The demand that all modern dwellings get direct sunlight has given these housing areas a new and more interesting character. The open style of the building, with parallel blocks whose orientation is determined with reference to the sun- [long axis] east-west if there are through-apartments or north-south. The first building type is preferred as it allows cross-ventilation and ensures that there is a side that is genuinely sunny. But it requires through-apartments which, not only reduce the depth of the building but also lead to longer facades as well as fewer apartments on each stairwell. This results in a system which is economically inferior to blocks that run from north to south"* (Acceptera, 1931).**

Figure 12: Home Insuranc Building, photo by unknown

Meanwhile, in Eastern Europe, modern architects started to design and build integrated residential buildings for the working class. However, the high cost of elevators and pressurised pumps led to early modern houses having multiple floor structures (Anthony Denzer, 2013).

To create the first high-rise buildings, Le Corbusier combined the concept and theory of "The Separation of Spaces of Function through Zoning" by Tony Garnier, skyscraper, and collective housing. As a result of the development of automobiles, planes, and especially high-rise housing, he developed utopian visions of the city of the future "The City To-morrow and its Planning" (Le Corbusier, 1925) and "Ville Radieuse" (Le Corbusier, 1933). In the books, the housing concept was explained, and the books have been influential on the architect. Through a series of documents such as the Congrès Internationaux d'Architecture Moderne (CIAM) and Charte d'Athènes, these theories and practices formed the idea

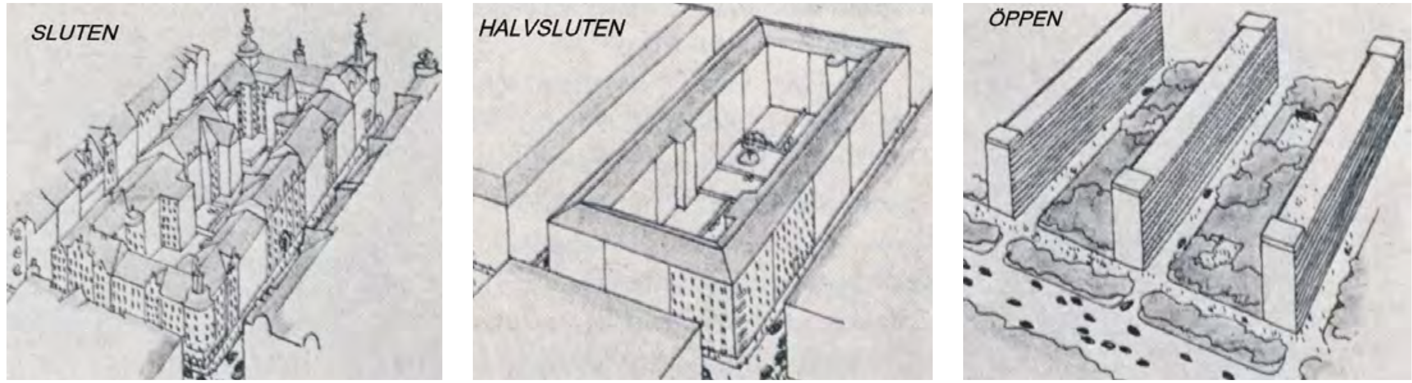
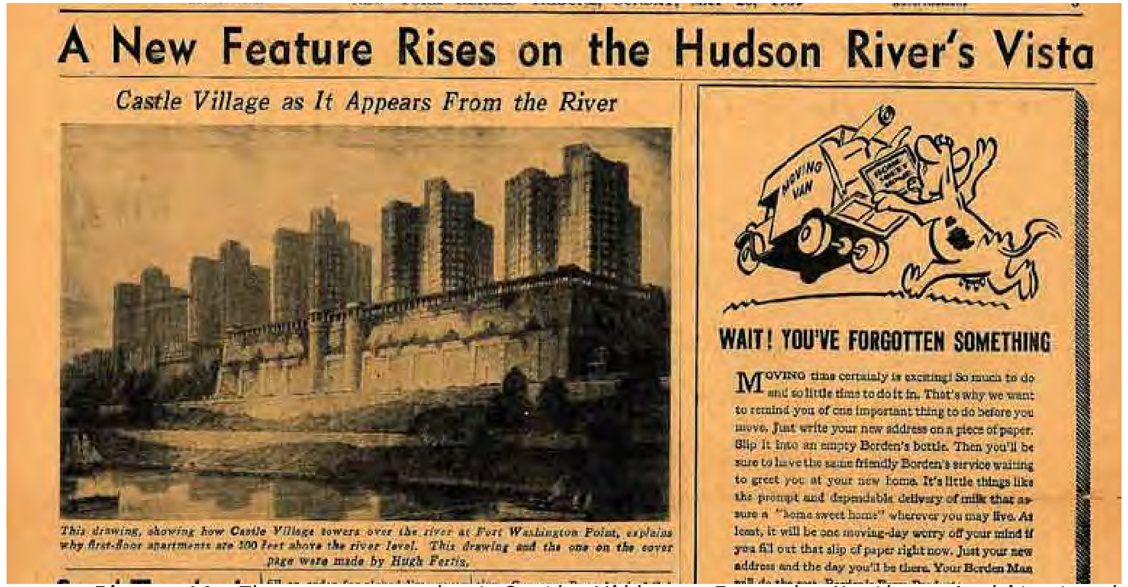


Figure 13: The development from the old closed urban planning system to the new open one: closed, semi-closed and open residential quarter, from Acceptera (Stockholm 1931).



A New Feature Rises on the Hudson River's Vista

Castle Village as It Appears From the River

This drawing, showing how Castle Village towers over the river at Fort Washington Point, explains why first-floor apartments are 100 feet above the river level. This drawing and the one on the cover page were made by Hugh Ferris.

WAIT! YOU'VE FORGOTTEN SOMETHING

MOVING time certainly is exciting! So much to do and so little time to do it in. That's why we want to remind you of one important thing to do before you move. Just write your new address on a piece of paper. Slip it into an empty Borden's bottle. Then you'll be sure to have the same friendly Borden's service waiting to greet you at your new home. It's little things like the prompt and dependable delivery of milk that assure a "home sweet home" wherever you may live. At least, it will be one moving-day worry off your mind if you fill out that slip of paper right now. Just your new address and the day you'll be there. Your Borden Man

Figure 14: The newspaper about Castle Village, Image by Northern Architectural Systems, unknown

of a "functional city" including modern high-rise housing and were widely accepted by architects, planners and government officials throughout Europe and the United States. By improving housing, environmental conditions, and employment opportunities, they claim that a new egalitarian society can be built and that high-rise housing is "the ultimate solution to the housing problem" (Stephen, Jane, Ignaz, 2008).

The development of high-rise housing shifted from Europe to the United States after the outbreak of the Second World War. The high-rise housing evolved from American skyscrapers, which differed from the parallel layout pattern of slab high-rises in Europe. The "Castle Village" is one of the earliest examples located in the middle of downtown New York. The high-rise apartment complex built in 1939 has five buildings composed of cooperative apartments, and each building has 14 floors and takes on the form of a "cross" (Northern Architectural Systems, 2022). High-rise housing practice and diffusion were affected by the Second World War, but the post-war housing development and construction of modern forms based on industrialisation, rationality and functionalism dominated.

Emergence: The post-war housing crisis leads to an utopian housing solution

After World War II, a new wave of rural migration to cities driven by job opportunities, family re-organisation and childbirth, as well as a wave of immigrants from colonial independence, forced governments to find a quick, efficient, and low-cost housing solution to meet the demand. Hereafter, public housing has become the main model for urban housing supply (Josh Lee, 2017). According to policymakers, the dominant function of the city represents a much more effective, equal, and fair society.

The high-rise residential houses can build a social structure close to the neighbourhood and create an environment for residents to interact and exchange ideas, a symbol of urbanisation and modernisation. By taking advantage of these ideas, governments and investors offer financial and policy support for residential high-rise development. After World War II, a new wave of rural migration to cities driven by job opportunities, family re-organisation and childbirth, and a wave of immigrants from colonial independence forced governments to find a quick, efficient, and low-cost housing solution to meet the demand. Hereafter, public housing has become the main model for urban housing supply (Josh Lee, 2017). According to policymakers, the dominant function of the city represents

Figure 15: Park Hill Estate, Image by Architectural Press Archive / RIBA Collections, unknown

a much more effective, equal, and fair society. The high-rise residential houses can build a social structure close to the neighbourhood and create an environment for residents to interact and exchange ideas, a symbol of urbanisation and modernisation. Taking advantage of these ideas, governments and investors offer great financial and policy support for residential high-rise development. High-rise housing has become a utopian housing solution due to this top-down development method.

In addition, several factors also play a role. First, technological innovation enabled the construction of high-rise residential buildings, cast-in-situ concrete technology, the use of large prefabricated homes, and a more rational production process for building in the construction field to improve the feasibility of residential buildings. In addition, high-rise residential buildings provide many amenities, including hot and cold water supply, central heating systems, recycling systems, etc. It attracts residents from all levels of society - they believe that high-rise residential living is a symbol of a modern lifestyle and a better quality of life. Many researchers have compared high-rise residential buildings with other forms of housing land usage. They note that residential buildings have many advantages, including solving the density issues, improved construction and sustainability, and spatial efficiency. (FEDERAL MINISTRY FOR ECONOMIC AFFAIRS AND CLIMATE ACTION, 2022). In summary, factors such as housing shortages, government aid, technological advancement, land saving, and modernism encouraged the development of high-rise residential areas throughout the world between the 1960s to 1980s from North America, Europe, Australia, or several developed regions such as Hong Kong and Singapore (Hans & Neil, 1992).

High-rise housing of this period has the following characteristics: As a predominantly public housing project, high-rise condominium residential developments received the largest amount of government subsidies during this period, and in Europe and the United States, multinationals took care of more than 80% of all affordable housing development (Tracy Kaufman, 2003). The precast concrete panel is assembled instead of bricks and mortar. Residential units adopt uniform standards, reinforcing economies of scale through repeated construction, and even the street width and block size become dependent upon the

Figure 16: The La Ville Radieuse, image by Le Corbusier, (1930).

crane technical index. Among the three is a remote location. The majority of the high-rise residential buildings are located at the edge of the city, easy to extend outward along the street, accordinga to the construction of a replication of the same pattern. It is a typical case study called Bijlmermeer. It was built in Amsterdam and established in 1975 as a public housing project that contains hundreds of residents and a hexagonal top board with a honeycomb effect (Tower Renewal Partnership, 2008).

Figure 17: Revisioning Amsterdam Bijlmermeer, Image by ASTADSARCHIEF AMSTERDAM, 1975

Decay: A stigmatised form of housing rejected by residents

Figure 18: The dilapidated of high rise apartment, South Africa.Posted by u/achmadSZN, 2019

The popularity of high-rise housing peaked during the 1960s and 1970s, but in the late 1970s, it had quickly become considered uninhabitable and a form of an urban slum. Today, urban slums are rapidly becoming uninhabitable and a source of concentrated poverty." The researchers have described a series of problems in high-rise residential after studying the European nations after World War II (Robert Gifford, 2000):

1. The structures are usually made without being tested because of poor quality construction methods and materials, such as asbestos, isolation, etc, and it caused a lack of sound insulation, dampness, condensation, and improper ventilation causing poor sound quality.

2. Insufficient equipment: insufficient heating, poor sanitation, inadequate storage space, a lack of elevators, public facilities, and insufficient external space.

2. Disadvantages of urban design and space: isolated location, high building density, traffic inconvenience, and noise pollution.

4. There are social problems in some geographic cities associated with noise, anti-social behaviour, crime, and insecurity, as well as poor relationships with your neighbours, especially if you live in a low-income community.

5. High rent and service charges for tenants; large operating losses on the landlord's part due to rent in arrears, high vacancy rate, and high maintenance costs.

6. The low status of the low image of the property makes it hard to compete with other housing types on the market.

7. Lack of resources, management confusion, and neglect are the reasons for management and organisational problems.

8. Intricacies in the legislation regarding the ownership of homes, blocks, and spaces around them.

9. A wide range of social and economic problems, including residents with high unemployment, low levels of education, cases of drug addiction, and similar family problems.

As a result of these issues, the high-rise residential living environments deteriorated rapidly. The newly built as a cleaning plan "slums" high-rise residential houses quickly became the new slums, some of which are still there today, while others have been gradually removed. During this period, all countries in Europe and the United States started to stop the construction of new high-rise residential buildings that had lower floors and more stories.

Figure 19: Ponte City Johannesburg, South Africa. 2008. Copyright Mikhael Subotzky and Patrick Waterhouse courtesy Goodman Gallery

Similarly, the Ponte Tower building in Johannesburg in 1975 is a 173-metre tall cylinder - still the tallest apartment building in Africa. The shopping mall, cinema, and even a bowling alley were designed to look futuristic even by today's standards. The residents could enjoy leisure activities without leaving the city since it was considered a "village". However, after the end of apartheid, the region's rising crime rates caused buildings to be gradually occupied by street gangs, which turned them into the world's highest and most magnificent slums (Mikhael Subotzky, 2008). But nothing is absolute, regarding Asia, Hong Kong, Singapore, and so on, where there is little land and a large population of the city, the high-level review after the war of high density of public housing development mode began to pay close attention to the residential end-users, experience, and evaluation, to explore methods to enhance the quality of the high-rise residential environment.

"Ponte sums up all the hope, all the wrong ideas of modernism, all the decay, all the craziness of the city. It is a symbolic building, a sort of white whale, it is concrete fear, the tower of Babel, and yet it is strangely beautiful."

- Norman Ohler "Stadt des goldes"

Regeneration: sustainable housing solutions for urban areas with high densities

By the end of the 1980s, the concept of sustainable development emerged, and the whole society began to reflect on the past development mode, especially a series of new theories for urban expansion were put forward, including 'New urbanism', 'Smart growth', 'Compact city' etc and improving the intensity of urban development and construction density has almost become the core content of every theory (Heba Allah Essam E. Khalil, 2010). According to the researcher, compact development has many advantages, including "reducing automobile dependence, reducing greenhouse gas emissions, lowering energy consumption, improving public transportation, revitalizing existing urban areas, protecting green spaces, and promoting commercial and trading activities." As a compact form of housing, the high-rise has been accepted in urban areas to provide sustainable housing development (Nazanin & Pirilsa, 2017).

Apart from the economisation of land use, residential buildings can also operate at higher energy efficiency and consumption of fewer resources. Comparing high-rise residential construction with other types of housing, high-rise residential requires fewer construction materials such as building foundations, roofs, and walls than other types. High-rise buildings have a greater potential for energy saving because their

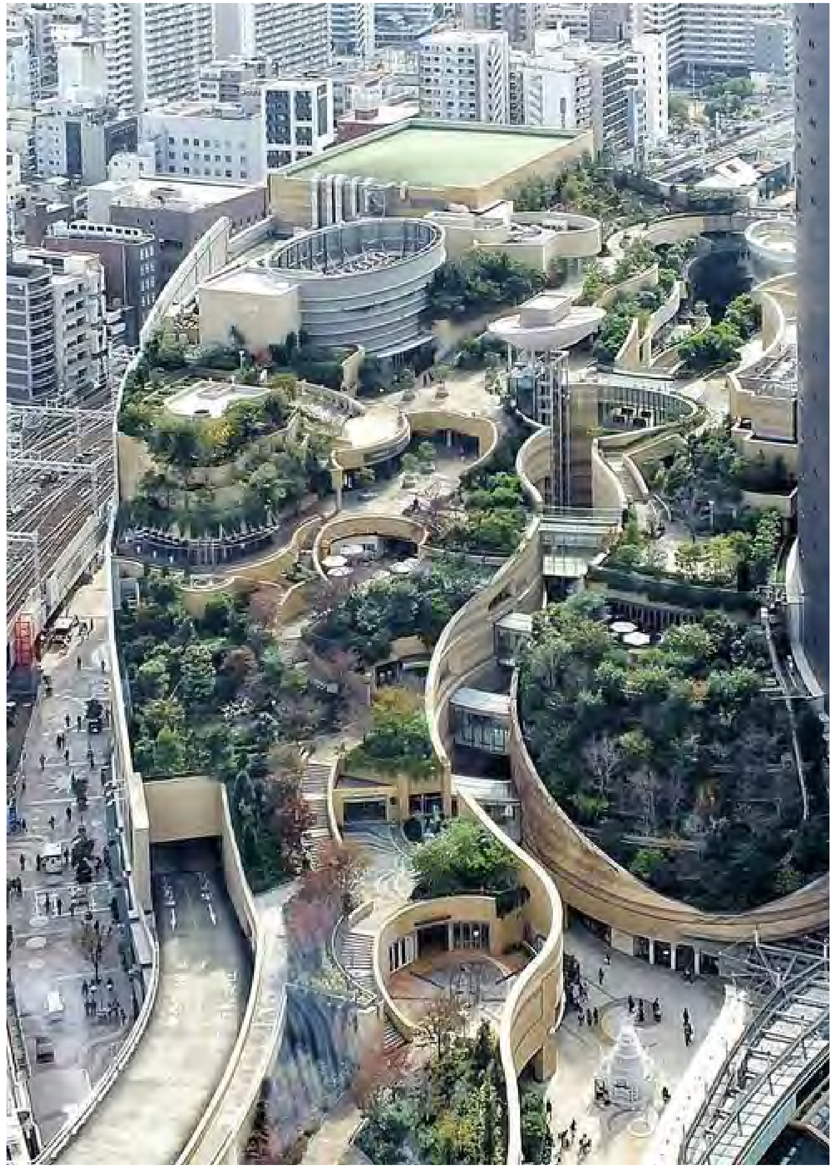


Figure 20: As an urban sprawl area, Japan's Namba Parks promotes diversity, complexity, and resilience with nearby urban housing. 2014. Copyright Yuka Yoneda

outer walls are smaller, they are located on middle floors, and they save on heating costs by reducing the ground, and the heat loss from the roof. High-rise buildings are also more efficient in terms of waste materials, garbage collection, recycling, and sewage disposal in addition to promoting the reduction of transportation energy consumption, since the use of public transportation are more common in high-rise residential complexes (Hazem, Khaled, Ezzat, Tarek, 2015).

Moreover, one of the advantages of high-rise residential is the concentration of all kinds of services and facilities within walking distance to meet the different types of family requirements. Additionally, the city encouraged residents to walk, ride a bike and utilise green travel routes to provide access to resources independent to all residents (children, adolescents, the elderly, people with disabilities, and those without vehicles). Higher densities also mean more housing for a larger population, more opportunities for working with others, shorter infrastructure standards such as cable, pipes and sewage systems, fewer routes for buses and roads, and, of course, larger facilities such as shops, hospitals, and schools, which improve efficiency and reduce costs.

However, there are also negative factors that affect the sustainability of high-rise residential development. A high-rise residential development will add to the burden on public services and infrastructure systems in a city. Increasing density results in a greater demand for supporting services within a unit area, which leads to more intense competition for limited land. Land for these facilities is often insufficient due to competition and occupation, resulting in a shortage of land. Additionally, when compared with low-rise buildings with embedded energy, the construction of high-rise residential buildings requires more equipment and accessories from a whole life cycle perspective.

Typically, the energy consumption of high-rise residential buildings is higher, since they typically have additional facilities such as elevators, pressurised water supply systems, air conditioning systems, etc. Demolition costs for high-rise residential buildings are higher and cause damage to land, groundwater, and other environmental resources irreversibly. Additionally, high-rise residences may not be conducive to a happy living environment. There are close ties between high-rise residential buildings and the urban heat island effect and the wind tunnel effect, both of which can negatively impact indoor and outdoor living environments (Peter, Thijs, Viktor, Bert, Jan, 2012). Higher

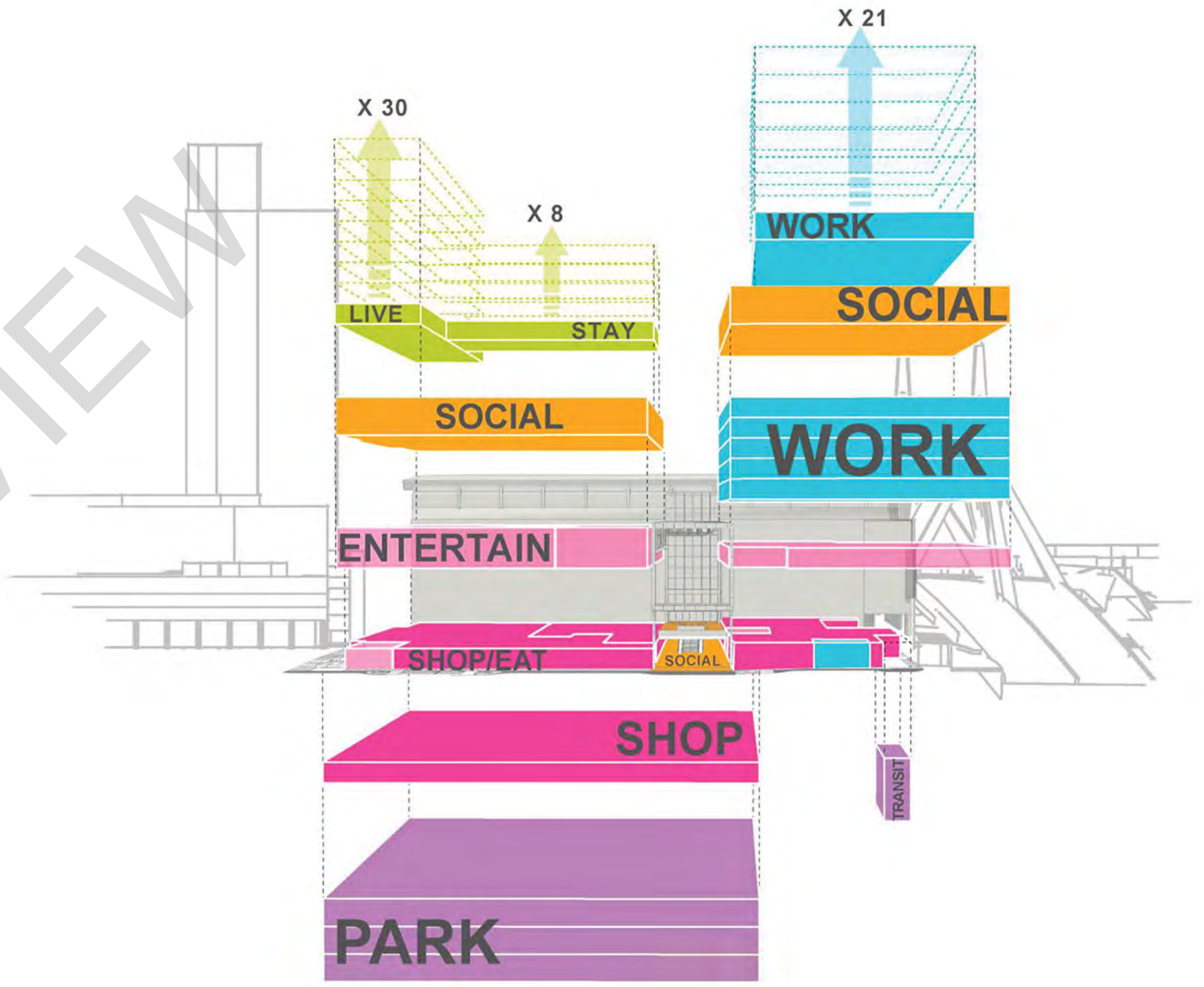


Figure 21: A diagram of The Hub on Causeway in Boston demonstrates how a mix of uses plug in to the podium, image by J.F. Finn, III and Duncan Paterson, 2020

buildings will affect natural lighting and ventilation in themselves and surrounding buildings, as well as cause environmental problems such as light pollution.

In response to these factors, professionals and enterprises across the globe have carried out a lot of research to find technical solutions, including strict planning controls to ensure the allocation of supporting facilities, research and development of new materials and processes to reduce embedded energy consumption, solar and wind energy to reduce operating energy consumption, vertical greening, and roof greening for environmental improvement, etc. Additionally, with the globalisation of the economy since the 1990s, a swell in urban populations and migration continued throughout the developed countries, helping the cities to slowly revive, resulting in a swell in investment and residential demands. (reference) As urbanisation advanced rapidly in developing countries, many people built their own homes at a rapid rate, laying the foundation for the rise of high-rise residential buildings. (reference) Through long-term continuous development and improvement, Asian cities represented by Hong Kong and Singapore have implemented high-rise residential development models adapted to local conditions, proving to the world high-rise resides is a feasible notion. Throughout the world, high-rise apartments quickly became a modern trend (Mir & Kheir, 2012).

Solution Of High-rise Liveability

There are certain benefits to living in high-rise apartment buildings. In contrast to the findings of many studies conducted in Europe and America, research from Singapore found that building separation between high-rise residential buildings provides residents with more open space, with a dramatic effect on transitioning from public, semi-public, semi-private to private, and providing more privacy, freedom, and space to move about, which fosters social connections between residents (Mir &Kheir, 2012).

Furthermore, the large-scale, well-maintained public green spaces between high-rises can provide residents of dense urban areas with high-quality, well-maintained outdoor spaces around their homes, which are very attractive to people living in those areas. There are other benefits of high-rise housing, such as the best views and quiet environment, as well as the sense of superiority and status that accompany them. Among high-rise residents, the residents benefit from panoramic views and a sense of elevation, and more female residents were drawn to the views while more male residents were enticed by the feeling of elevation.

A study analysed the data and found high-rise home prices and values correlated with their views and floor heights. One of the perceived benefits of renting apartments on higher floors of tall buildings is that they have unobstructed views. As evidenced by surveys of residents, apartments with appealing views are more expensive. Residents above the fifth floor experienced lower traffic-related airborne pollution during the non-heating season, resulting in enhanced air quality. High-floor apartments are more silent - residents will have better sleep quality, auditory discrimination, and reading performance than low-floor apartments (Fan Ng, 2017). As a result of high-rise living, certain groups of people find that lifestyle appealing, like young white-collar workers, couples without children, empty-nesters, and people with high incomes.

Some groups of people are more likely to be satisfied with high-rise living as opposed to other types of living arrangements, according to residential satisfaction studies. It is necessary to expand the concept of a residential area beyond merely the building in which people live, to include the surrounding area as well as the people who live there. Housing quality should continue to improve for every resident. An ideal resident is satisfied with their living situation. To put it another way, the residents' goal is hedonism (AYDOAN, 2005). As a result, childless couples and young single adults were more comfortable and satisfied with high-rise living. This had to do with the fact that they preferred the benefits of high-rise living i.e., socialising, urban lifestyle, etc.

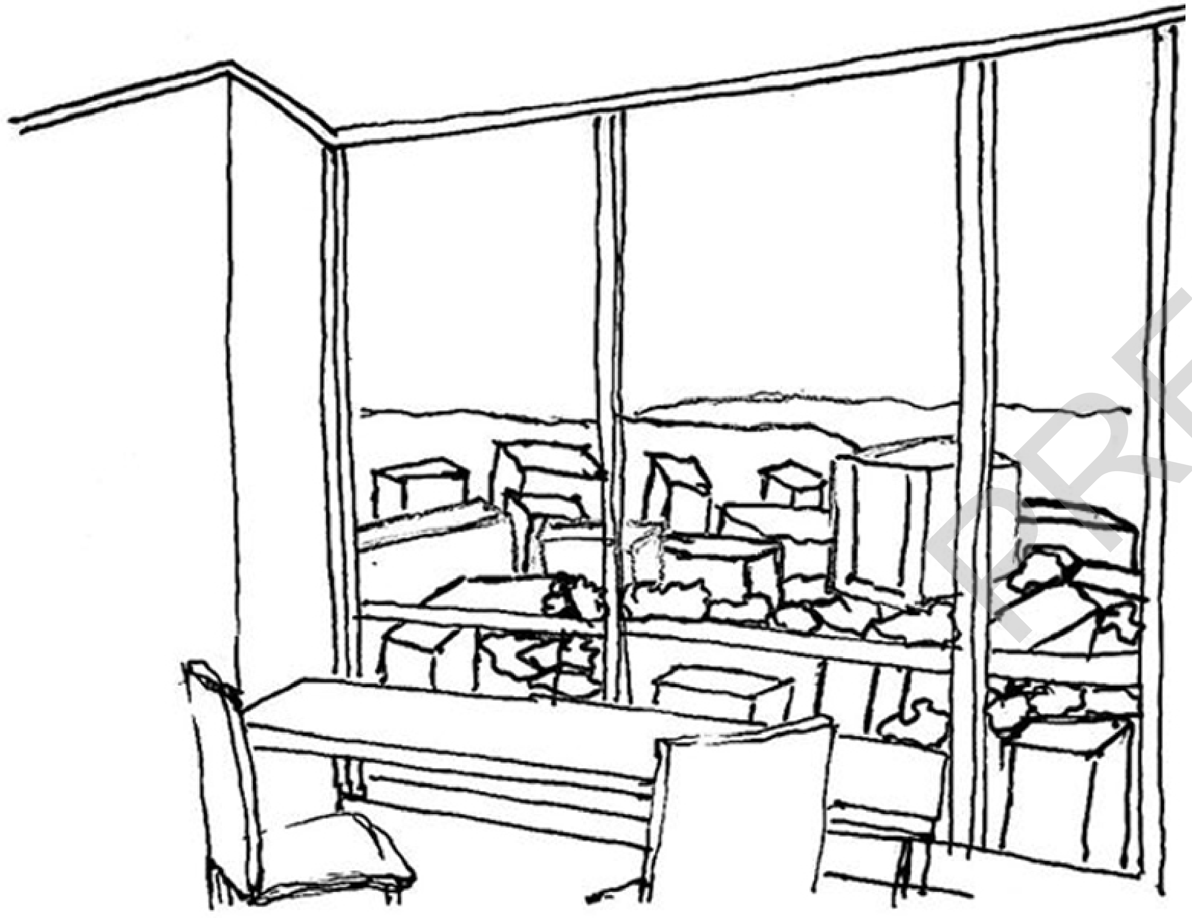


Figure 22: The sketch of view the city at high rise units, drawing by Zhihao Zhang

Liveability Improvements in High-rise Housing

Energy is consumed during the construction of high-rise buildings, environmental harmonization is experienced, and people's quality of life is negatively affected, as well as community development. In addition to their impact on the growth and development of communities, high-rise buildings have also been extensively studied for their energy consumption and environmental impact (Belinda and Anthony, 2010).

In addition to studying these technologies' sustainability, energy consumption patterns, design, and technology, several studies were conducted. Many questions must be addressed by residents, including accessibility, landscaping, security, and air quality. Positive correlations were found between participation levels and levels of satisfaction.

To ensure the social sustainability of a community, sustainable housing becomes an important component of a high-rise building. The amenities provided by an integrated community contribute to its success. Densification, the process of mixing uses and increasing density, has its supporters, but it can also lead to isolation and social isolation, which can lead to social withdrawal (Barger, 2016). The architect has to contend with the challenges of social interaction and community feeling when designing high-rise living for families with children.

An architect should explore the possibility of high rises generating their power, reducing energy consumption and protecting the environment are closely linked. It is important for designers to carefully select the glazing systems for their buildings to ensure a building's energy performance. Apart from the windows, partitions and external walls should also be considered. When the design phase is completed, embodied energy should be reduced. Thermal insulation reduces building heating and cooling costs, according to research on design and technology. Natural ventilation is an important part of sustainable environmental practices (Chela & Kaushikb, 2018). Sustainable high-rise buildings should take into account all these factors to enhance residents' quality of life.

Sustainability is a concept that architects relate building to the enhanced quality of life. There are two different sustainability that could develop the quality of high-rise building life: environment sustainability and social sustainability. Community quality and social interaction between residents make up social sustainability. Residents' physical conditions can be improved by environmental sustainability.

In this respect, a comfortable living environment is not limited to building design, but also involves other aspects of sustainability. An example is a connection between urban sustainability and high-rise living. Creating sustainable cities involves balancing residential development, environmental protection, employment, housing, basic services, social infrastructure, and transportation equity while ensuring equity in income, employment, housing, and basic necessities. However, researchers have shifted their attention to neighbourhoods and districts opposed to individual buildings, allowing them to assess the built environment, public transportation, and services (Gabrielle, 2017), etc. Although individual high-rise responses to sustainability can lead to incremental improvements on a larger scale, this may require late-stage improvements in building design.

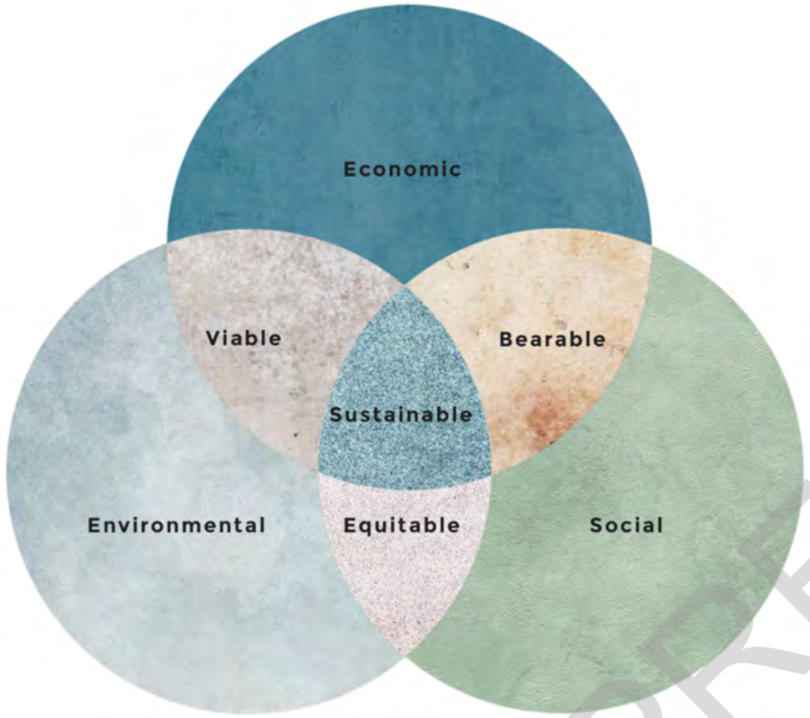


Figure 23: Triple bottom line of sustainability for high rise building concept, drawing by Zhihao Zhang

Developing a social sustainability high-rise

Over the past few decades, social sustainability has been increasingly influencing urban policy, housing, and the planning of cities worldwide - recently, social sustainability has gained increased attention as a fundamental component of sustainable development. Social sustainability and urban development are often in close association consequently, representing the development of physical spaces that enhance the quality of life of the users. Research indicates that spatial qualities were made by reflecting these needs in building space design - Comfort, Privacy, Legibility, Diversity, Public Participation, Visual Richness, Sense of Place and Identity (Kheir, 2011).

High-rise developments are important when it comes to residents' liveability. Comfort should be considered in the design process at the earliest stages. Having a better understanding of well-being has led residents in the high-rise to seek improvements in living conditions. For the liveability concept to work, it is crucial to correlate the physical environment and resident satisfaction (Kostas, 2022). During the design process, if the conditions and requirements of the building are optimised to create a more comfortable living environment, fewer problems may arise, and more solutions present themselves. Regarding physical planning, future condo developments should incorporate recreational areas and enhance residents' quality of life through a pleasant social environment. Rather than simply serving as a shelter, housing should be assessed regularly and actively to create a liveable and appealing environment. There is no arbitrariness concerning community spaces, and resident patterns and needs must be taken into account when they are reorganized. Many options can be considered. For example, facilities such as outdoor rest areas, gyms, and renovations or expansions of existing facilities could be considered. As well as incorporating study rooms and book clubs into the living program, places with little use but high demand are needed. In addition, places will be designed more effectively if they are designed according to generation or age.

Residents' satisfaction is influenced by two variables: the objective features of the built environment and moderators, which are not related to the environment (Cristina, 2019). As part of a causal link between the environment and outcomes, these moderators contribute to differences in outcomes. The macro-context and individual demographic characteristics of residents are two groups of moderators identified in different studies. Researchers have found that demographic factors like gender, life stage, and income level moderate residents' perceptions of high-rise housing in many studies. However, climate, housing systems, and diversity of housing types have not yet been studied in evaluating high-rise buildings. Some research studies have found that buildings with attractive entrances and lobby areas reduce communication between neighbours. Furthermore, buildings with more security have less beautiful landscaping, and buildings with more security talk less about one another's problems. The inhabitants of buildings with pleasant green spaces trust each other less. These facts are responded to in the design of the buildings.

An important aspect noted throughout the research is that although the interior design and decoration of homes are closely related to housing quality, there are no institutional measures in place to ensure it (Yaolin, 2022). Inadequate policies are the result of the lack of effective procedures for gathering user feedback and opinions. Quality of life is increasingly important in planning and design, and user voices must be heard. Important questions are: Does it keep unwanted influences such as noise and pollution out of the building; does it provide privacy, both visual and acoustic; and does it ensure a safe personal space? Finally, questions can be asked whether

it provides a comfortable thermal and visual environment for working, resting, and other daily activities, and whether it provides a healthy environment: fresh air and hygiene. (NG and Wong, 2003).

The paper contains some concrete results that help to further shape the process of sustainable communities in Orange Grove (Johannesburg) by proposing a conceptual model for future high-rise projects. Residents are more satisfied with their lives when they have access to attractive community spaces. From the correlation analysis between housing satisfaction and housing program, residents are more satisfied when they have access to common community spaces and programs and participate in the community. Residents see the building as the foundation for a vibrant, resilient, and sustainable community where everyone's needs are met and community life is rich.

As shown in the study above, a better community area with amenities, such as lounges, can enhance the relationship between residents who live in different units, thereby socially sustaining the relationship between the residents. The provision of spaces like lounges, green roofs, dining rooms, game rooms, lobbies, pools, and fitness centres can encourage social interaction. Building amenities could include balcony spaces,

Figure 24: Communication and social gathering are encouraged within the NION Tower's interior, image by UNStudio

hallways, communal spaces, and outdoor spaces around the building as well as across multiple levels.

When these spaces are conceived collectively and planned in a connection, the quality of the amenity spaces will be enhanced by recognizing their significance of them and their ability to function together. As a result, a sense of ownership and participation emerges in shared spaces to foster a sense of belonging to those spaces. Especially when high-rise buildings are not equipped with comfortable green spaces, a rooftop terrace is highly recommended. In addition, buildings designed for less mobile citizens should have green spaces that are physically and visually accessible.

For high-rise housing to be socially sustainable, it must take into account the ideas and expectations of its residents. As a means of enhancing the demand for high-rise housing, steps should be taken to ensure that the same favourite features that are found in low-rise neighbourhoods are also found in high-rise buildings, as well as that they are designed to meet the needs of a diverse range of users.

When mixed-use is combined with residential, office and commercial space, the streetscape can be activated.